Lab Exercise 9- Create Service in Kubernetes

Name-Manish
SAP ID- 500119723
Batch-2

Objective:

- Understand the syntax and structure of a Kubernetes Service definition file (YAML).

Prerequisites

- Kubernetes Cluster: Have a running Kubernetes cluster (locally using Minikube or kind, or a cloud-based service).
- kubectl: Install and configure kubectl to interact with your Kubernetes cluster.
- Basic Knowledge of YAML: Familiarity with YAML format will be helpful for understanding Kubernetes resource definitions.

Step-by-Step Guide

NodePort Service

To expose the Service on a port on each Node in the cluster, modify the Service type to NodePort.

Create a YAML file named ***service.yaml*** with the following content:

service.yaml

```
apiVersion: v1 kind:
Service metadata:
  name: nodeport-service
spec:   selector:     app:
web   ports:
    - protocol: TCP       port: 80       targetPort: 80       nodePort:
30007 # A specific port in the range 30000-32767   type:
NodePort
```

Explanation:

- The primary difference from the ClusterIP Service is the addition of nodePort, which specifies the static port on each Node.
- type: Set to NodePort, exposing the Service on a specific port across all Nodes.

Apply this YAML to create the NodePort Service:

```
kubectl apply -f nodeport-service.yaml
```

```
PS D:\kubs> kubectl apply -f nodeport-service.yaml
error: the path "nodeport-service.yaml" does not exist
PS D:\kubs> kubectl apply -f service.yaml
service/nodeport-service created
PS D:\kubs> kubectl get services
```

Verify the Service:

kubectl get services

```
service/nodeport-service created
PS D:\kubs> kubectl get services
NAME                TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1      <none>        443/TCP        15m
nodeport-service    NodePort    10.96.196.18   <none>        80:30007/TCP   10s
PS D:\kubs>
```

You should see the nodeport-service listed with a NodePort and details about the port exposed.